

Section 4

2026年1月27日 星期二 12:35

4 EEP1 ALU Instructions

Task 4. Use ISSIE's waveform simulator to simulate the first eight clock cycles of the instructions in lab1testmovadd. Use both the assembly code and the waveform simulation to answer the questions below.

lab1testmovadd.txt

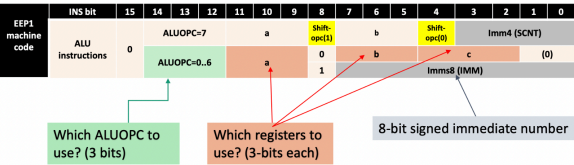
```
MOV R5, #3 // set up known values on R5...R7, test MOV with literal op2
MOV R6, #-1
MOV R7, #13
MOV R8, R6 // test MOV
ADD R1, R5, R7 // test ADD
ADD R2, #3 // test ADD with literal op2
ADD R4, #255 // test ADD with literal op2
```

Assembly Code

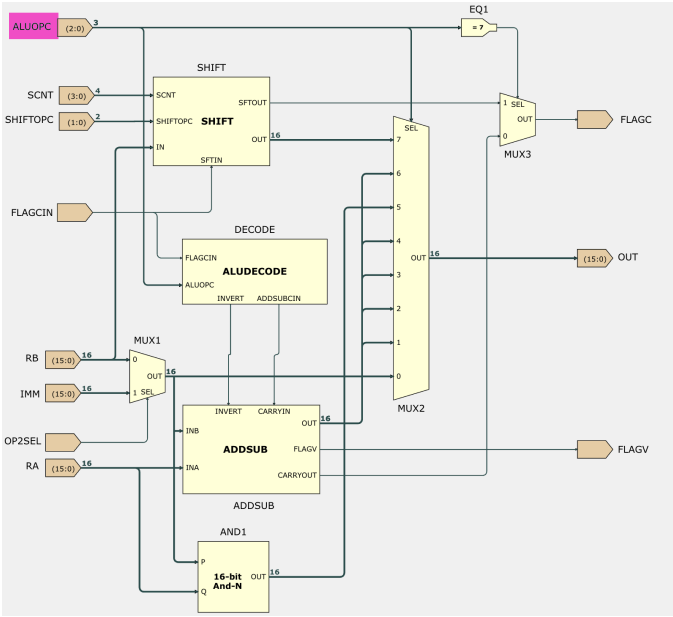
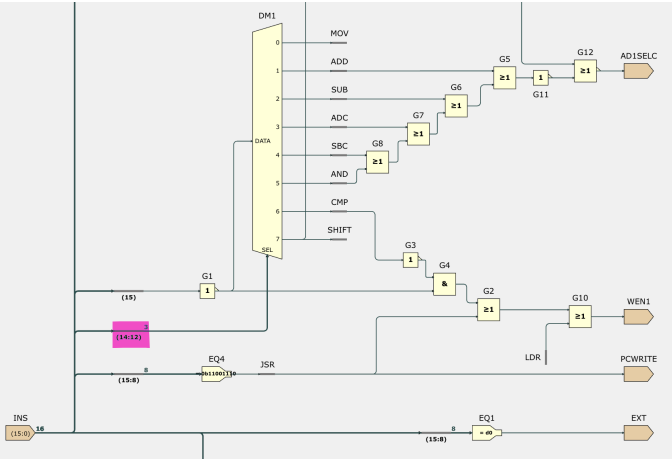
	0	1	2	3	4	5	6	7	8	9	0
Reg16x8.R7(15:0)	x0000		x0000	x000D						x000D	x0000
Reg16x8.R6(15:0)	x0000		xFFFF							xFFFF	x0000
Reg16x8.R5(15:0)	x0000		x0003							x0003	x0000
Reg16x8.R4(15:0)	x0000						x0000	xFFFF		xFFFF	x0000
Reg16x8.R3(15:0)	x0000									x0000	x0000
Reg16x8.R2(15:0)	x0000					x0000	x0003			x0003	x0000
Reg16x8.R1(15:0)	x0000				x0000	x0010				x0010	x0000
Reg16x8.R0(15:0)	x0000			x0000	xFFFF					xFFFF	x0000

Simulation

1. Through which port does the ALU receive the information about which operation to perform (ADD, SUB, AND, etc.)? State which field(s) of the 16-bit instruction word are connected, directly or indirectly, to this port.



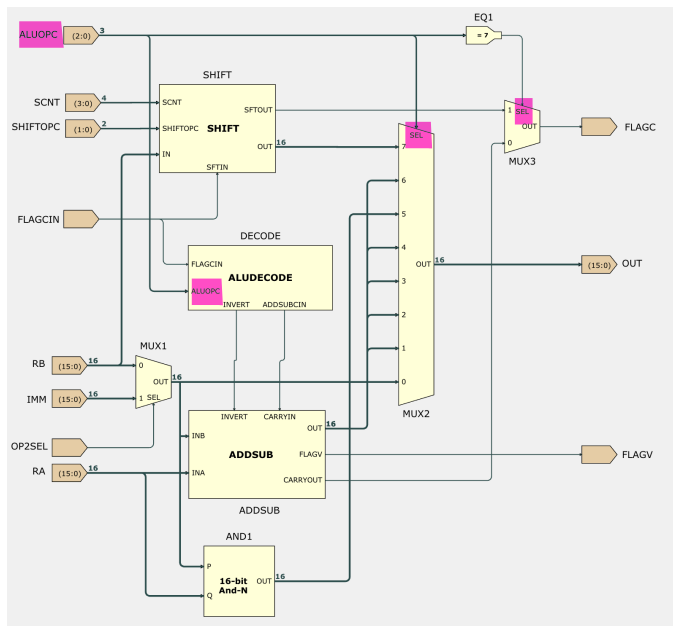
12-14 bits determine the operation.



	0	1	2	3	4	5	6	7	8	9	0
Datapath.DECODE.Aluopc(2:0)	0				0	1		1	0		0

And it imports to ALU through ALUOPC.

Then, using the hardware, explain how this port connects to the ALU.DECODE block and other internal ALU blocks (such as ADDSUB). You do not need to understand the detailed internal design of these blocks yet; you only need to explain how the instruction word selects one ALU operation rather than another.



Inside ALU, ALUOPC determines whether ADDSUB need carrier or invert, so it goes to ALU.DECODE. It also connects to MUX2 to choose which function to operate in the end.